

Total No. of Questions : 8]

SEAT No. :

P825

[Total No. of Pages : 2

[5870]-1148

T.E. (Information Technology)
COMPUTER NETWORK & SECURITY
(2019 Pattern) (Semester - II) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer four questions from the following.
- 2) Figyres to the right side indicate full marks.
- 3) Draw the neat sketches wherever required.

Q1) a) Explain with diagram layerd architecture & clustered architecture for sensor network. **[9]**

b) Comment on Adhoc Netwrok MAC layer with design issues, Design goal. **[9]**

OR

Q2) a) Explain the issues in designing a Routing protocol for adhoc wirless network. **[9]**

b) Explain different issues and challenges in designing sensor network. **[9]**

Q3) a) What is the importance & need for security and explain network attack. **[8]**

b) Explain different block cipher muscles. **[9]**

OR

Q4) a) Explain distributed denial of service attacks in details **[8]**

b) Explain with suitable examples what do you mean by active attacks & passive attacks. **[9]**

Q5) a) Write a short notes on **[9]**

i) Encryption

ii) Decryption

b) Explain RSA in details **[9]**

OR

P.T.O.

- Q6)** a) What is digital signature? What are application of digital signature [9]
b) Explain following terms. [9]
i) Cryptography
ii) Symmetric key Cryptography
iii) Asymmetric key Cryptography

- Q7)** a) Write short notes on [9]
i) Malware
ii) Phishing
iii) MITA attack
b) Elaborate different cyber security polycies in detail and explain different challenges in internet governance. [8]

OR

- Q8)** a) Explain the term cyber stalking. How to iderty and detect cyber stalking.
b) Explain the types of cyber-crimes.



Total No. of Questions : 8]

SEAT No. :

PA-1507

[Total No. of Pages : 2

[5926]:127

T.E. (Information Technology)
COMPUTER NETWORK & SECURITY
(2019 Pattern) (Semester-II) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3, or Q4, Q5 or Q6, and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain DSDV & AODV protocol in detail. **[9]**

b) Write a short note on sensor network with classification of protocols used. **[9]**

OR

Q2) a) Explain design issues and design goal in adhoc network MAC layer. **[6]**

b) Compare sensor network with adhoc wireless network. **[6]**

c) Explain with diagram clustered architecture for sensor network. **[6]**

Q3) a) Explain different block cipher modes. **[8]**

b) Comment on security principles and security services. **[9]**

OR

Q4) a) What is importance and need of security?. **[5]**

b) Explain distributed Denial of service attacks. **[6]**

c) Explain Electronic Code block (ECB) mode **[6]**

Q5) a) Explain difference between symmetric and asymmetric key cryptography. **[9]**

b) Explain Data Encryption Standards with diagram. **[9]**

OR

P.T.O.

- Q6)** a) Explain following terms: [9]
- i) Groups
 - ii) Rings
 - iii) Prime Numbers
- b) Explain diffie Hellman key exchange in detail. [9]

- Q7)** a) What is cyber terrorism? Write in detail example of cyber terrorism. [8]
- b) What is a man-in-the-middle attack (MIM)? Explain in detail. [9]

OR

- Q8)** a) Explain the term phishing & SQL Injection with suitable example. [8]
- b) Explain the term cyber stalking & cyber espionage with suitable example. [9]



Total No. of Questions : 8]

SEAT No. :

P489

[6003]-710

[Total No. of Pages : 2

T.E. (Information technology)
COMPUTER NETWORK AND SECURITY
(2019 Pattern) (Semester - II) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) All Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain MACAW protocol in details. **[9]**

b) Explain with diagram Layered Architecture for Sensor Network. **[9]**

OR

Q2) a) Explain the issues in designing a routing protocol for Ad-hoc Wireless Network. **[6]**

b) What are hidden station and exposed station problem in WLAN. **[6]**

c) Explain different issues and Challenges in Designing a Sensor Network. **[6]**

Q3) a) What is stream cipher? Explain encryption process using stream cipher with suitable example. **[8]**

b) What is Cipher Block Chaining (CBC)? Explain the process of CBC with suitable diagram. **[9]**

OR

Q4) a) Describe the following network security threats. **[5]**

i) Unauthorized access

ii) Distributed Denial of Service (DDoS) attacks

b) Describe the following fundamental principles of Information security **[6]**

i) Integrity

ii) Authentication

iii) Authorization and Access Control

c) What is Cipher Feedback Mode (CFB) and Electronic Code book (ECB)? **[6]**

P.T.O.

Q5) a) Explain Data Encryption Standard Algorithm in detail with suitable diagram. **[9]**

b) Explain Diffie-Hellman key exchange algorithm. **[9]**

OR

Q6) a) Explain Private Key Management. **[9]**

b) Explain following terms. **[9]**

- i) PKIX Model
- ii) Digital Signature
- iii) Digital Certificate

Q7) a) Write a short note on Software attacks & hardware attacks with example. **[8]**

b) Explain the threats and vulnerabilities of the information security system. **[9]**

OR

Q8) a) Explain Layers of Cyber Security in detail. **[8]**

b) What is a man-in-the-middle attack (MIM)? Explain in detail. **[9]**



Total No. of Questions : 8]

SEAT No. :

P-7625

[Total No. of Pages : 2

[6180]-145

T.E. (Information Technology)

COMPUTER NETWORKS AND SECURITY

(2019 Pattern) (Semester - II) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer the questions - Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of electronic calculator is allowed
- 5) Assume suitable data, if necessary.

Q1) a) Explain Adhoc Network MAC Layer with Design Issues, Design Goal & Classification. [9]

b) Classify protocols used in Sensor Network. [9]

OR

Q2) a) Explain AODV and DSR protocol in detail. [9]

b) Explain the issues in designing a Routing Protocol for Ad-hoc Wireless Networks. [9]

Q3) a) Explain the need & importance of security and types of attack. [8]

b) What are different security principles and security services? [9]

OR

Q4) a) Explain Block cipher modes in detail. [8]

b) Explain network security threats with example attacks. [9]

Q5) a) Define symmetric key encryption? Explain DES in detail. [9]

b) Elaborate Digital signatures, Digital Certificates. [9]

OR

P.T.O.

- Q6)** a) Write a note on RSA algorithm. [9]
b) What is private key? Explain Diffie-Hellman key exchange algorithm. [9]

- Q7)** a) What is cyber terrorism? How to identify and detect cyber stalking? [8]
b) Explain various cyber crimes and Computer criminals. [9]

OR

- Q8)** a) Define & explain Vulnerability, Threat, Malware & Phishing. [8]
b) Explain DoS Attack and SQL injection attack. [9]



Total No. of Questions : 8]

SEAT No. :

PB3865

[Total No. of Pages : 2

[6262] 129

T.E. (Information Technology Engineering)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester - II) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Explain the connection establishment and data transfer phase in the following routing protocols with suitable diagram. **[9]**

- i) AODV
- ii) DSDV

b) Explain with diagram Layered Architecture for Sensor Network. **[9]**

OR

Q2) a) Differentiate between infrastructure network and infrastructure less network. **[6]**

- b)** What are hidden station and exposed station problem in WLAN. **[6]**
- c)** Explain Distributed Denial of Service attacks. **[6]**

Q3) a) What is network attack? Write short note on following with suitable example? **[8]**

- i) Active attack
- ii) Passive attack

b) What is Cipher Feedback Mode (CFM)? Explain the process of CFM with suitable diagram. **[9]**

OR

P.T.O.

- Q4)** a) What is Electronic Code Book (ECB)? Explain the process of ECB with suitable diagram. [5]
- b) Describe the following network security threats [6]
- i) Unauthorized access
 - ii) Distributed Denial of Service (DDoS) attacks
- c) Differentiate between Active attack and Passive attack [6]

- Q5)** a) Define and list various computer network security mechanisms Also write short notes on the following terms [9]
- i) Encryption
 - ii) Decryption
- b) Define and explain digital signature. What are the applications of digital signature. [9]

OR

- Q6)** a) Define the following mathematical preliminaries and also explain the following in brief [9]
- i) Prime numbers
 - ii) Group
- b) What is Cryptography? also explain the following terms [9]
- i) Public key encryption
 - ii) Hash function

- Q7)** a) What is Cyber terrorism? What is an example of cyber terrorism? [8]
- b) Define Crime and Cybercrime. State Cybercrimes classification. [9]

OR

- Q8)** a) What Is Cyberstalking explain with an example. [8]
- b) Explain the term Phishing & SQL Injection with a suitable example. [9]

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Total No. of Questions : 8]

SEAT No. :

PC1808

[6353]-127

[Total No. of Pages :2

T.E. (Information Technology)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester- II) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or 2, Q.3 or 4, Q.5 or 6, Q.7 or 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Explain DSDV (Destination Sequenced Distance Vector) with connection establishment and data transfer phase in detail. **[9]**
b) Comment on Adhoc Network MAC Layer with Design Issues, Design Goal. **[9]**

OR

- Q2)** a) Briefly explain classification of routing protocol for AdHoc Wireless Networks. **[6]**
b) Explain with diagram Clustered Architecture of Sensor Network. **[6]**
c) Write a short note on MACAW. **[6]**

- Q3)** a) What is stream Cipher? Explain the encryption process using stream Cipher with suitable example. **[8]**
b) What is Cipher Feedback Mode(CFM)? Explain the process of CFM with suitable example. **[9]**

OR

- Q4)** a) Write comparison between symmetric and asymmetric key cryptography. **[5]**
b) Define Network Attack. Explain with suitable example what do you mean by Active attacks and Passive attacks? **[6]**
c) Describe the following network security threats: **[6]**
i) Unauthorized access
ii) Distributed denial of service
iii) Man in the middle

P.T.O.

- Q5) a)** Explain PKIX Model. [9]
b) Explain Advanced Encryption Standard with diagram. [9]

OR

- Q6) a)** Explain following terms [9]
i) Hash Function
ii) Digital Signature
iii) Digital Certificate
b) Explain the working of DES algorithm with suitable example. [9]
- Q7) a)** Define Crime and Cybercrime? State Cybercrimes classification. [8]
b) What is the difference between Information security and cyber security?
Explain the different layers of security? [9]

OR

- Q8) a)** Explain the concept of Software attacks & hardware attacks with example. [8]
b) Write a short note on: [9]
i) Malware
ii) Phishing
iii) Cyber Threat



Total No. of Questions : 8]

SEAT No. :

PD4329

[Total No. of Pages : 2

[6403]-127

T.E. (Information Technology)
COMPUTER NETWORKS & SECURITY
(2019 Pattern) (Semester - VI) (314451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Figures to the right indicate full marks.

- Q1)** a) Explain DSR in details with route discovery & route maintenance. [9]
b) Comment on Adhoc Network MAC layer with Design issues, Design Goal. [9]

OR

- Q2)** a) Briefly explain classification of routing protocol for Adhoc Wireless Network. [6]
b) Explain with diagram layered Architecture of Sensor Network. [6]
c) Write a short note on MACAW. [6]

- Q3)** a) What is stream cipher? Explain the encryption process using stream cipher with suitable example. [8]
b) What is Cipher Feedback Mode (CFM)? Explain the process of CFM with suitable example. [9]

OR

- Q4)** a) What is substitution cipher? Explain the mono alphabetic cipher and poly alphabetic cipher. [5]
b) Define Network Attack. Explain with suitable example what do you mean by active attack & passive attacks. [6]
c) Describe the following network security threats: [6]
i) Unauthorized access
ii) Distributed denial of service
iii) Man in the middle

P.T.O.

Q5) a) Explain private key management. [9]

b) Explain advanced encryption standard with diagram. [9]

OR

Q6) a) Explain following term : [9]

- i) Hash function
- ii) Digital signature
- iii) Digital certificate

b) Explain PKIX Model. [9]

Q7) a) Define crime and cybercrime? State cybercrimes classification. [8]

b) Define the term cyber security & explain layers of security. [9]

OR

Q8) a) Explain the concept of software attack & hardware attacks with example. [8]

b) Write short notes on : [9]

- i) Malware
- ii) Phishing
- iii) Cyber threat

